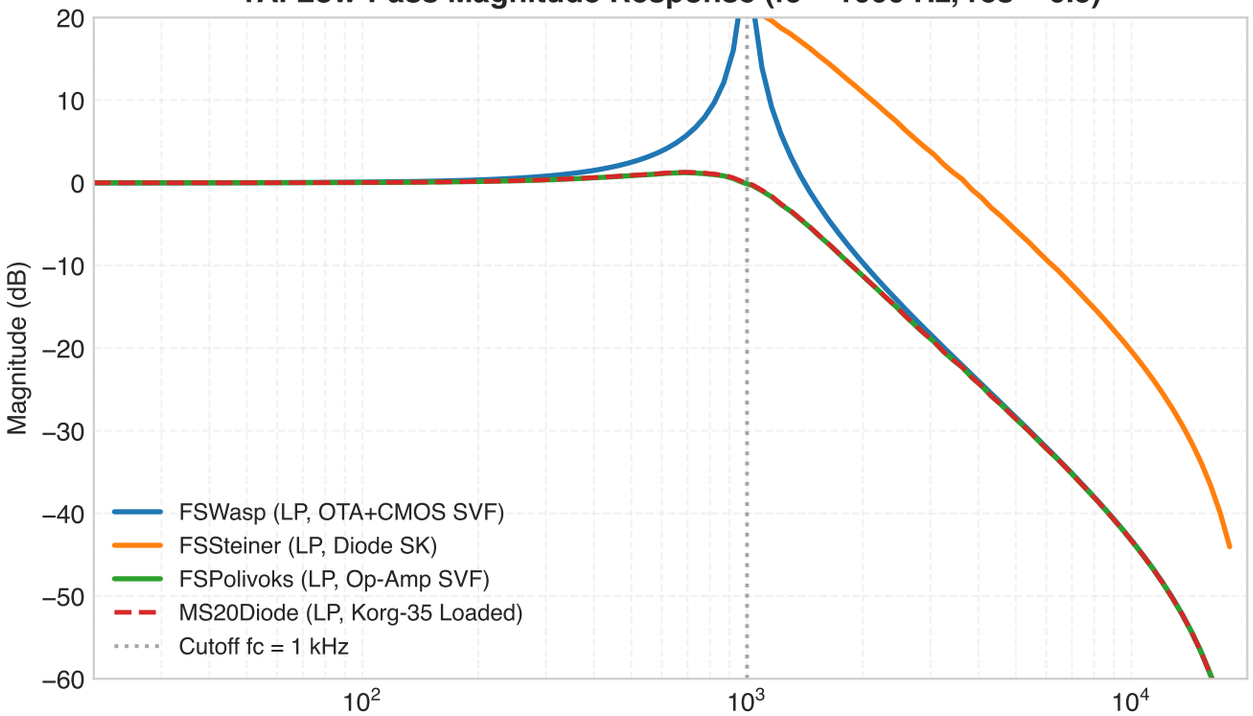
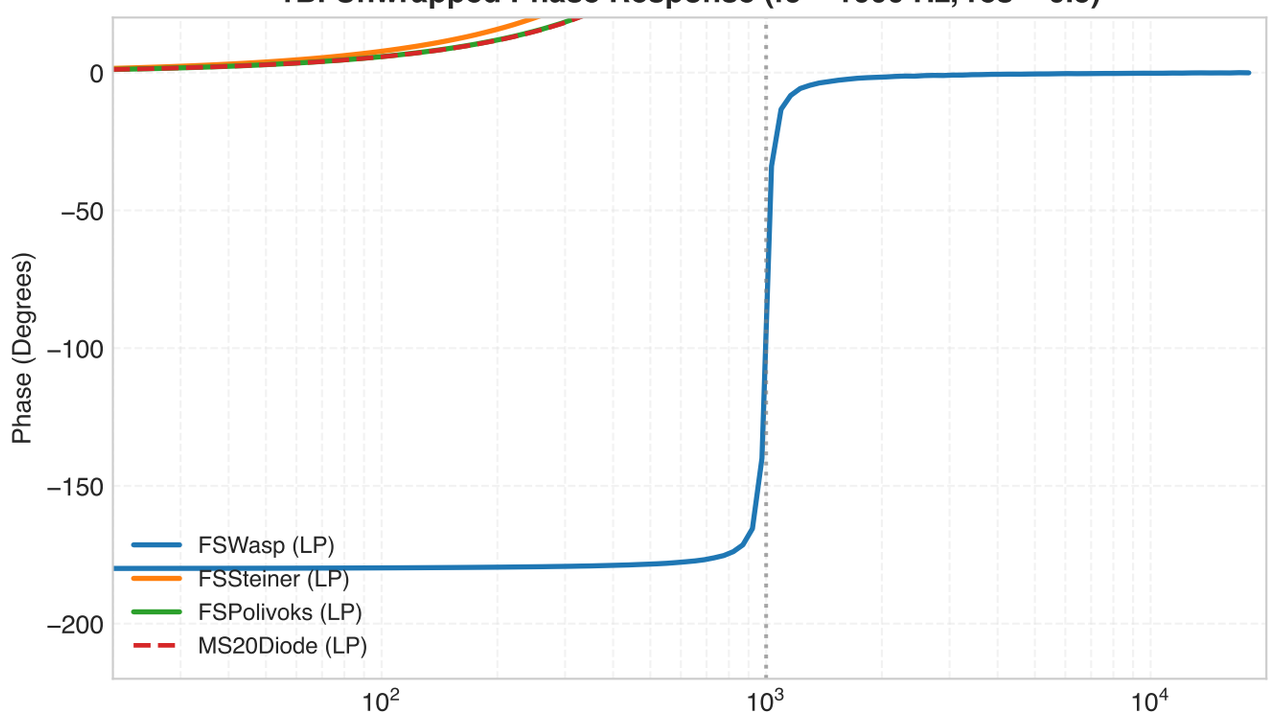
FilterSuite: Frequency & Multimode Phase Characterization (Small-Signal Linear)

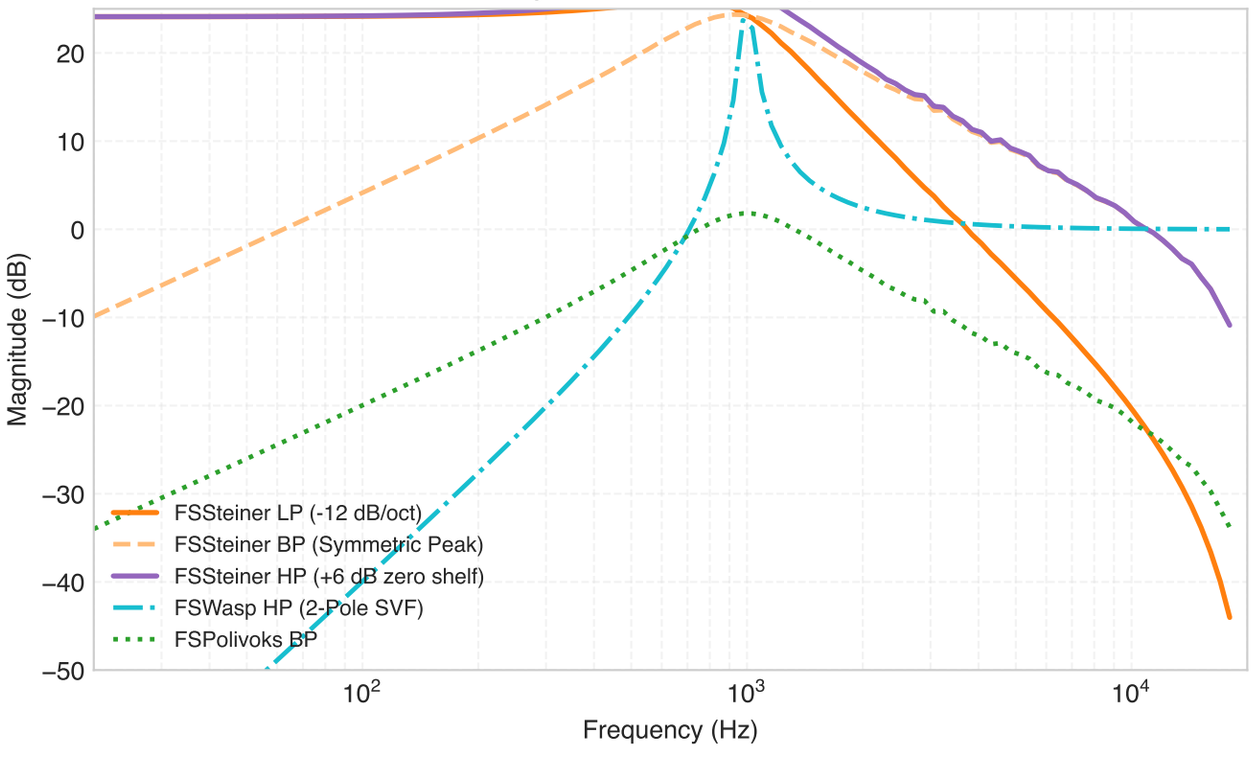
1A. Low-Pass Magnitude Response (fc = 1000 Hz, res = 0.5)



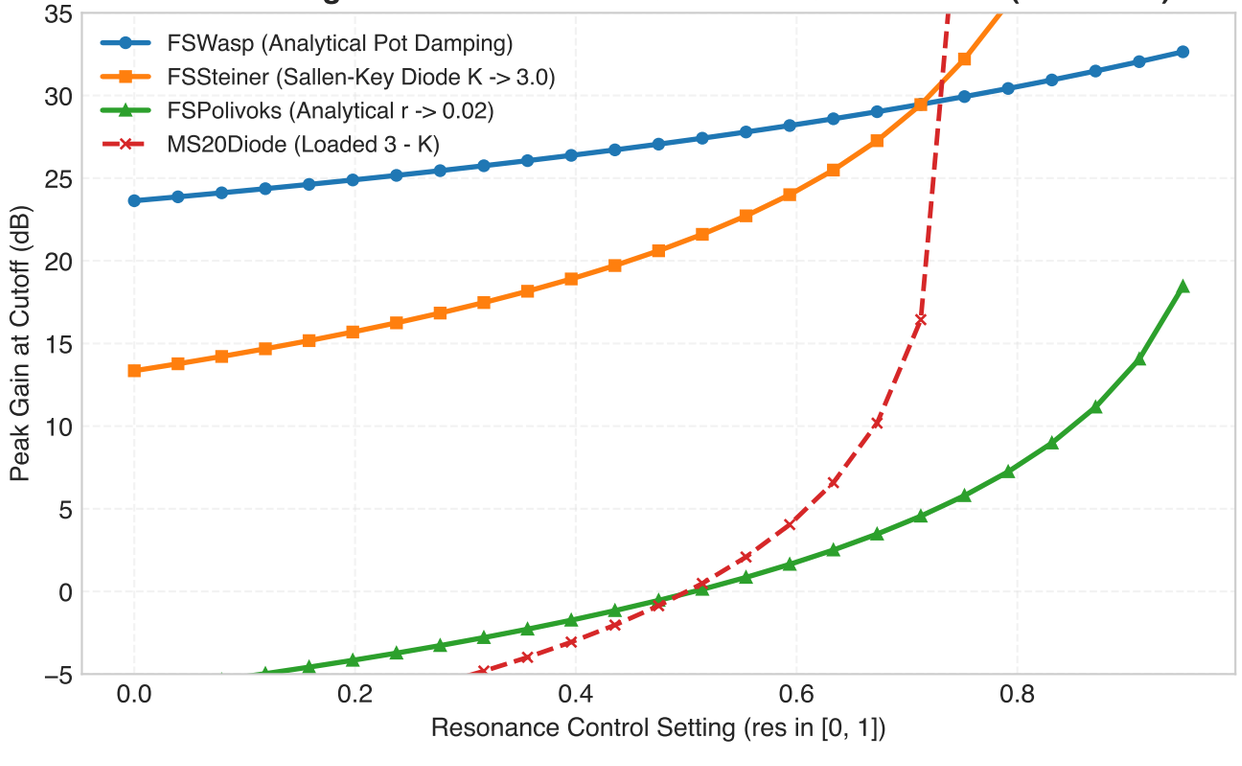
1B. Unwrapped Phase Response (fc = 1000 Hz, res = 0.5)



1C. Multimode Diversity: LP vs BP vs HP (fc = 1000 Hz, res = 0.6)

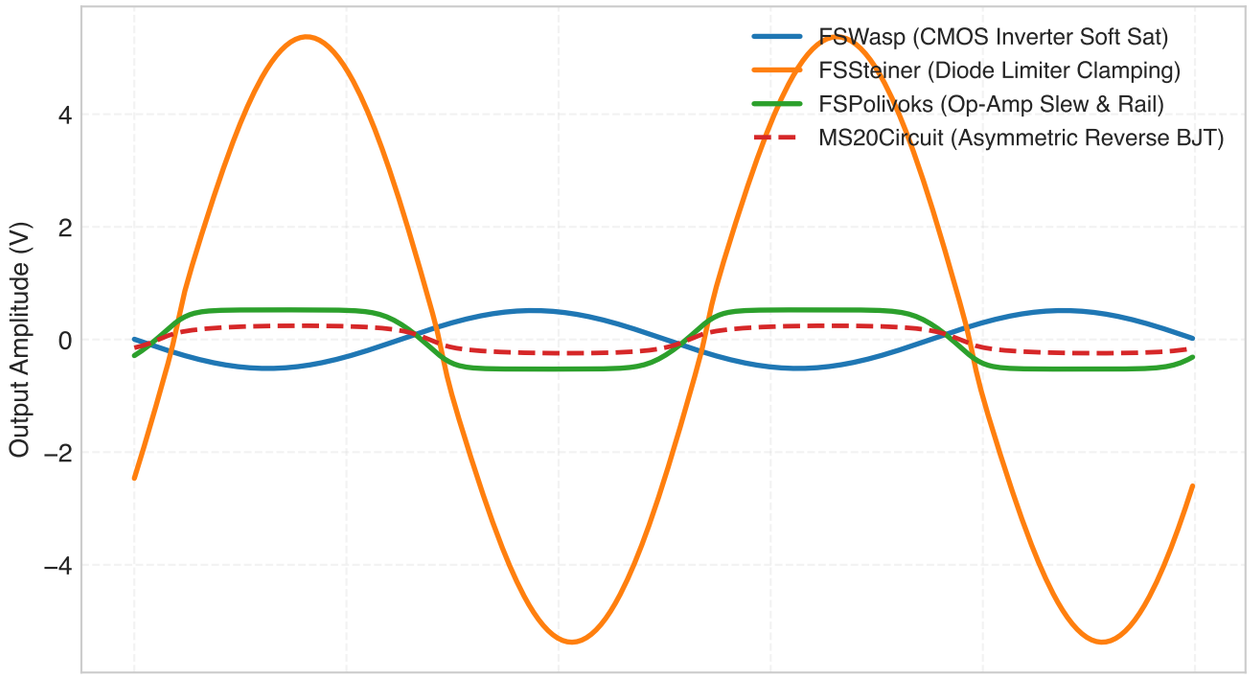


1D. Small-Signal Resonance Peak Growth vs Control Knob (fc = 1 kHz)

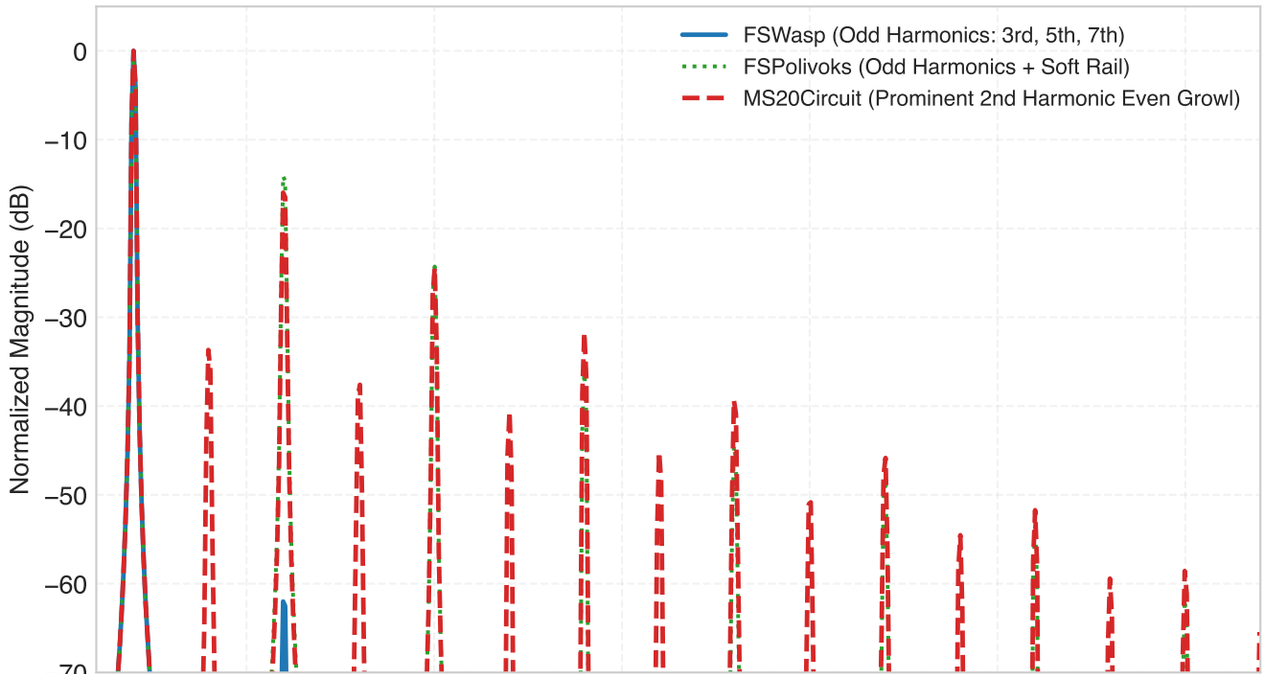


FilterSuite: Nonlinear Distortion, Harmonic Spectra & Waveform Shaping

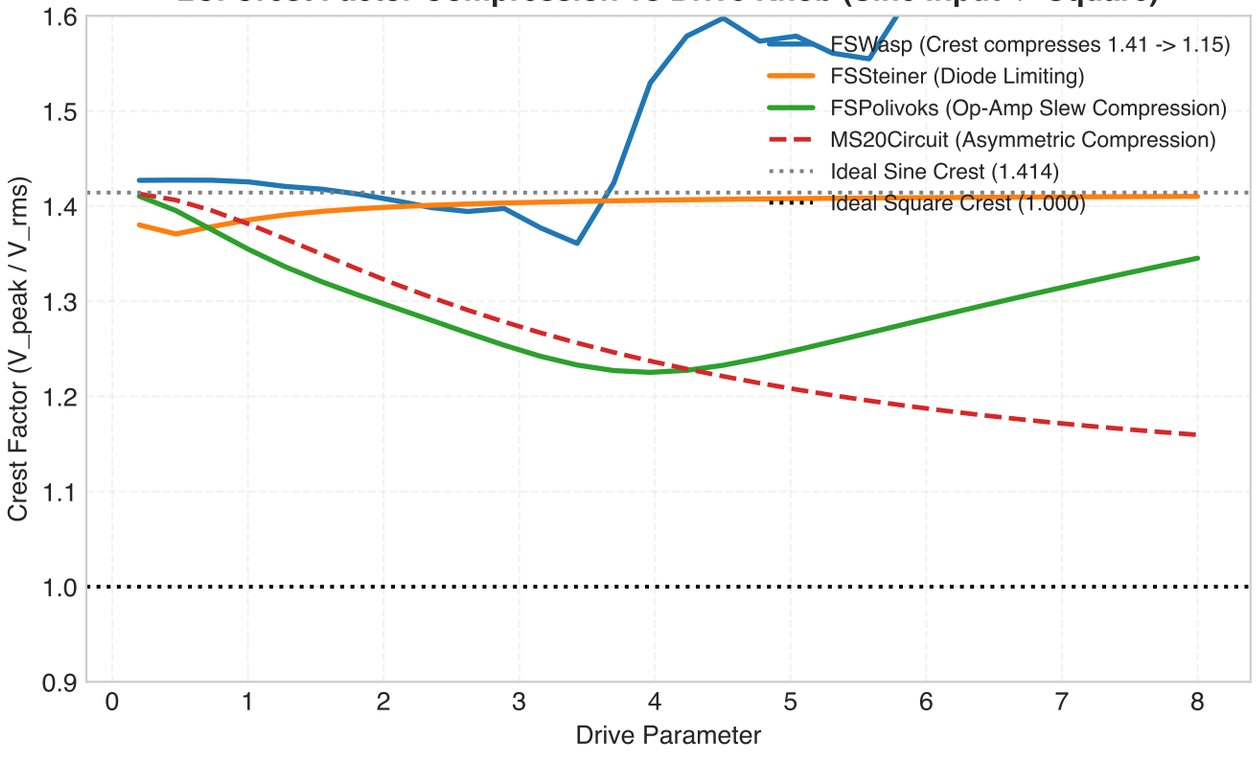
2A. Steady-State Waveforms (200 Hz Sine, drive = 3.5, fc = 1200 Hz)



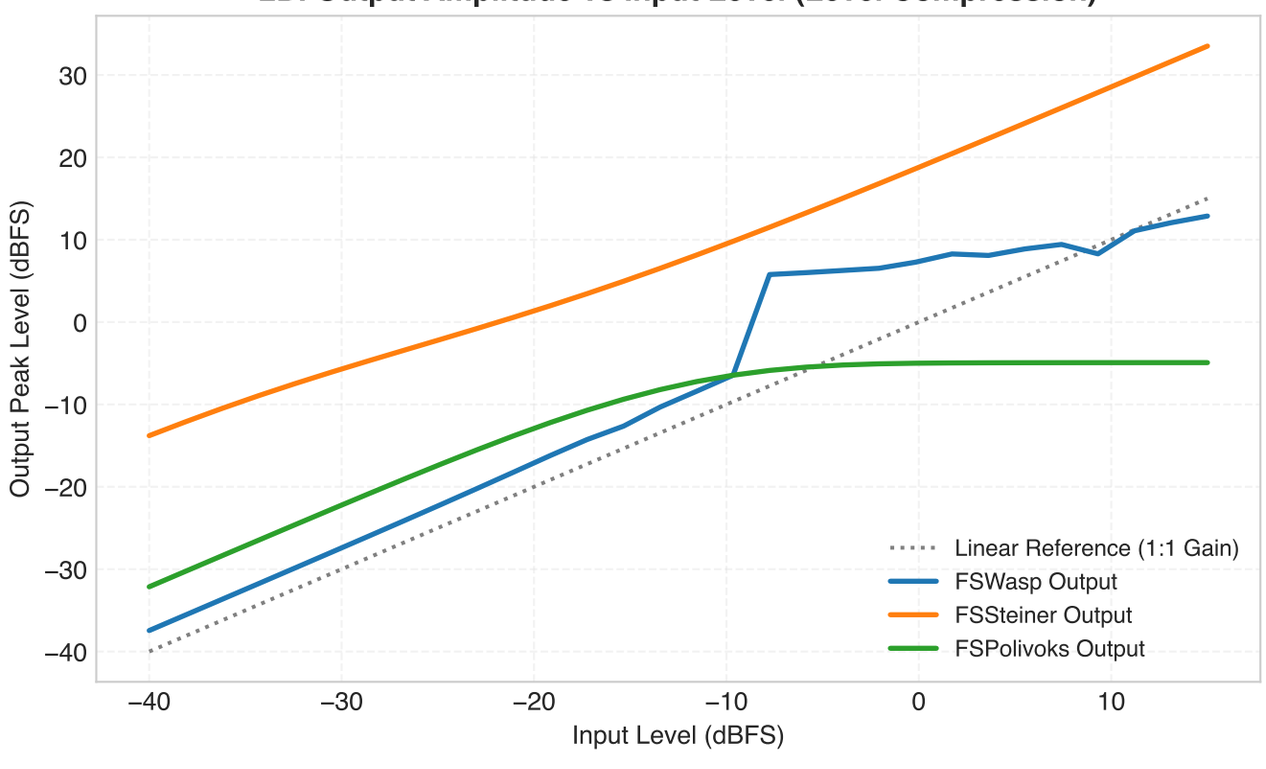
2B. Harmonic Distortion Spectrum (FFT up to 3 kHz)



2C. Crest Factor Compression vs Drive Knob (Sine Input -> Square)

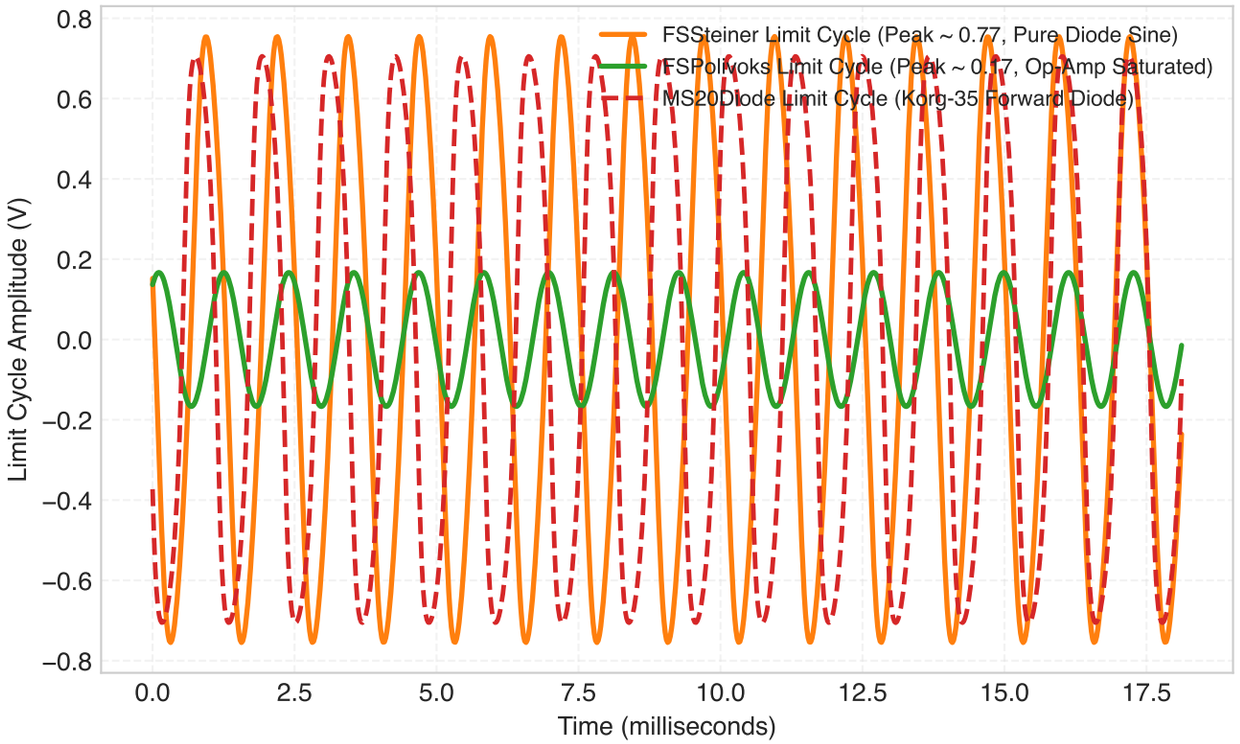


2D. Output Amplitude vs Input Level (Level Compression)

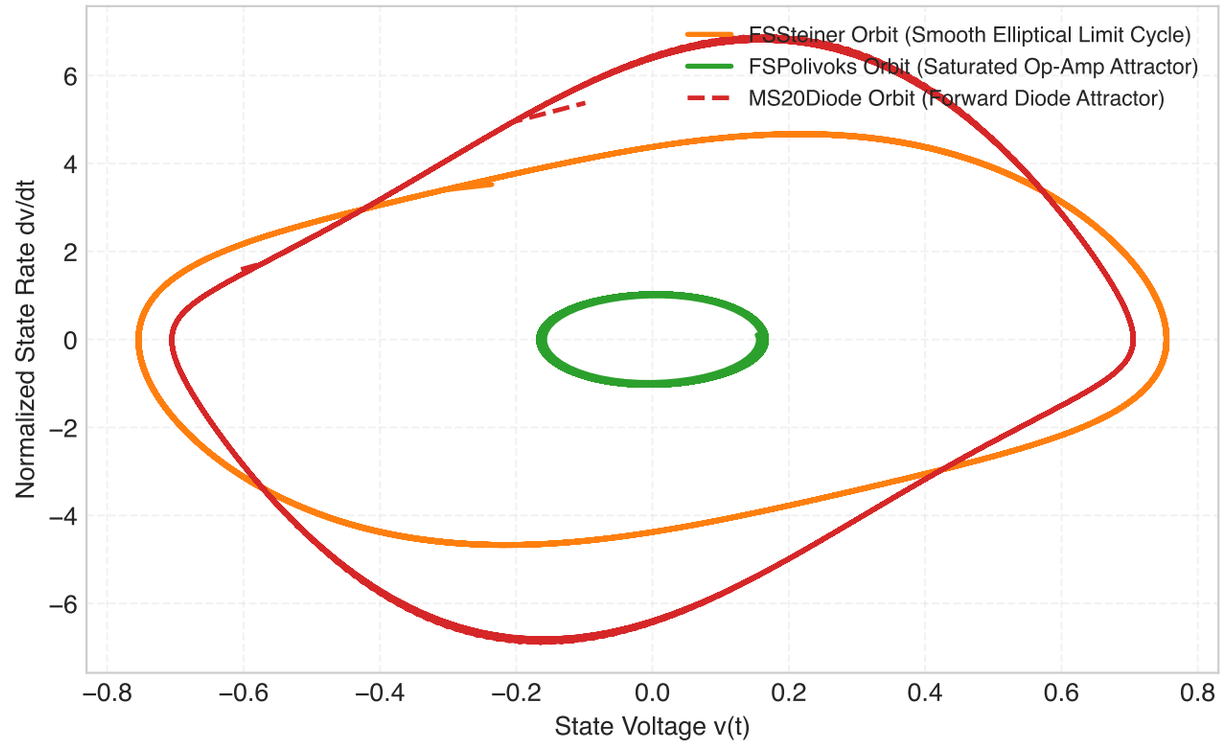


FilterSuite: Autonomous Self-Oscillation, Phase Plane & Audio FM

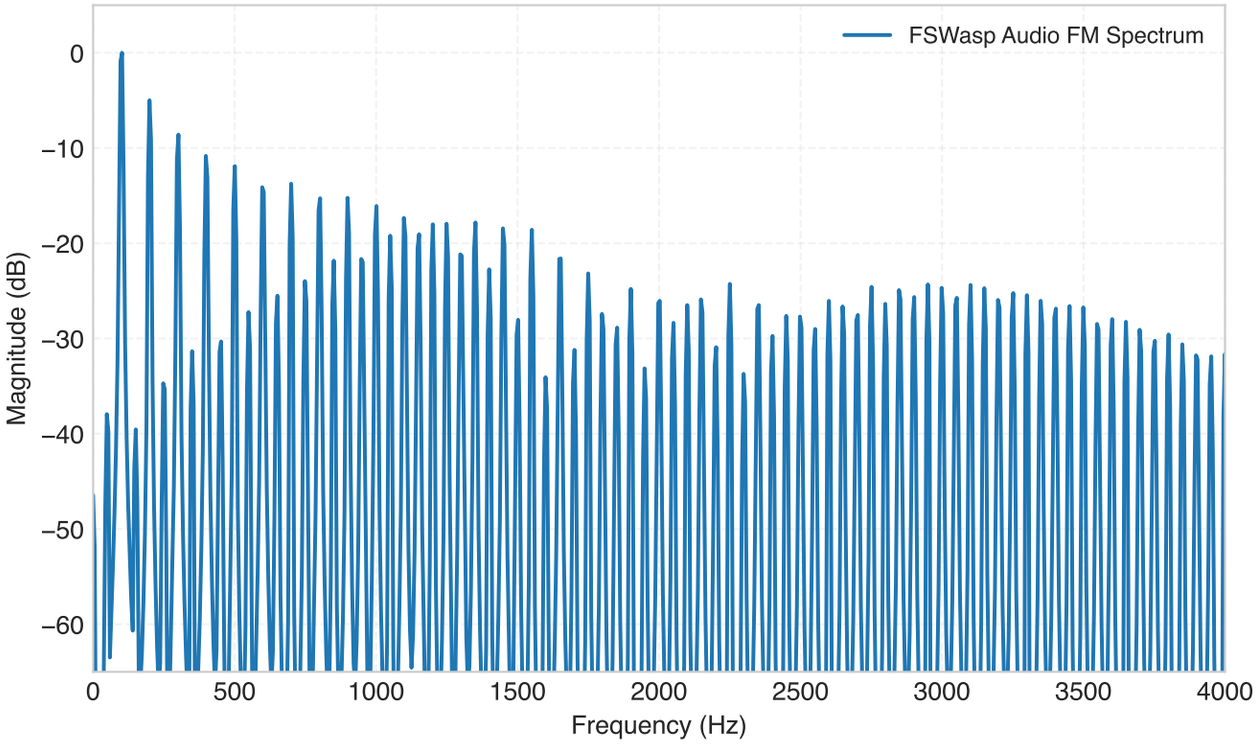
3A. Autonomous Self-Oscillation Limit Cycles (fc = 880 Hz, Zero Input)



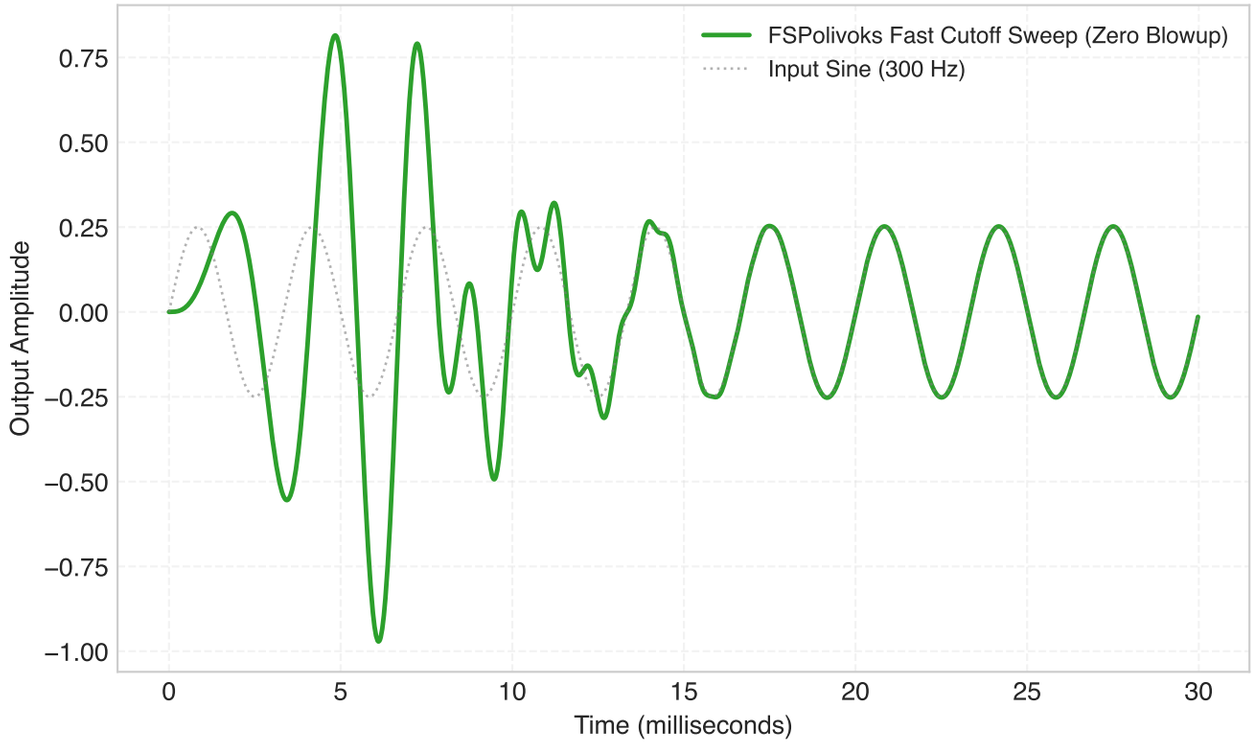
3B. State-Space Phase Plane Portrait (Limit Cycle Attractors)



3C. Audio-Rate Filter FM Spectrum (100 Hz Saw, 150 Hz Modulator)

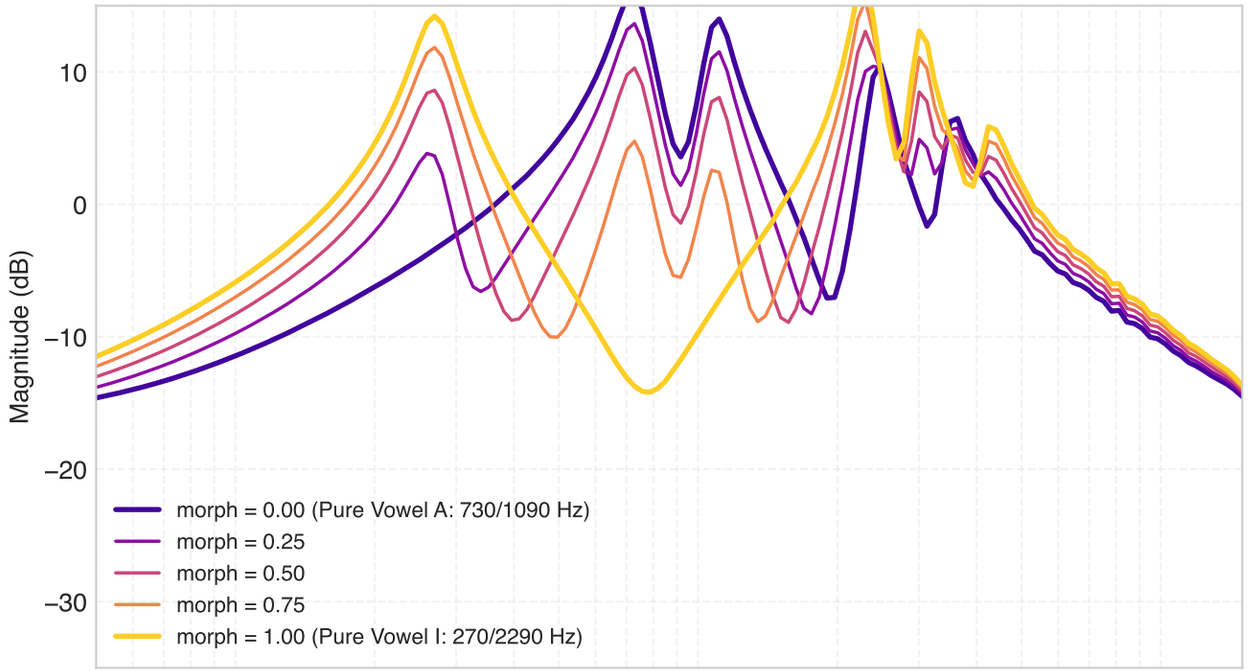


3D. Fast Cutoff Sweep (4 Octaves in 20 ms: 200 Hz -> 3200 Hz)

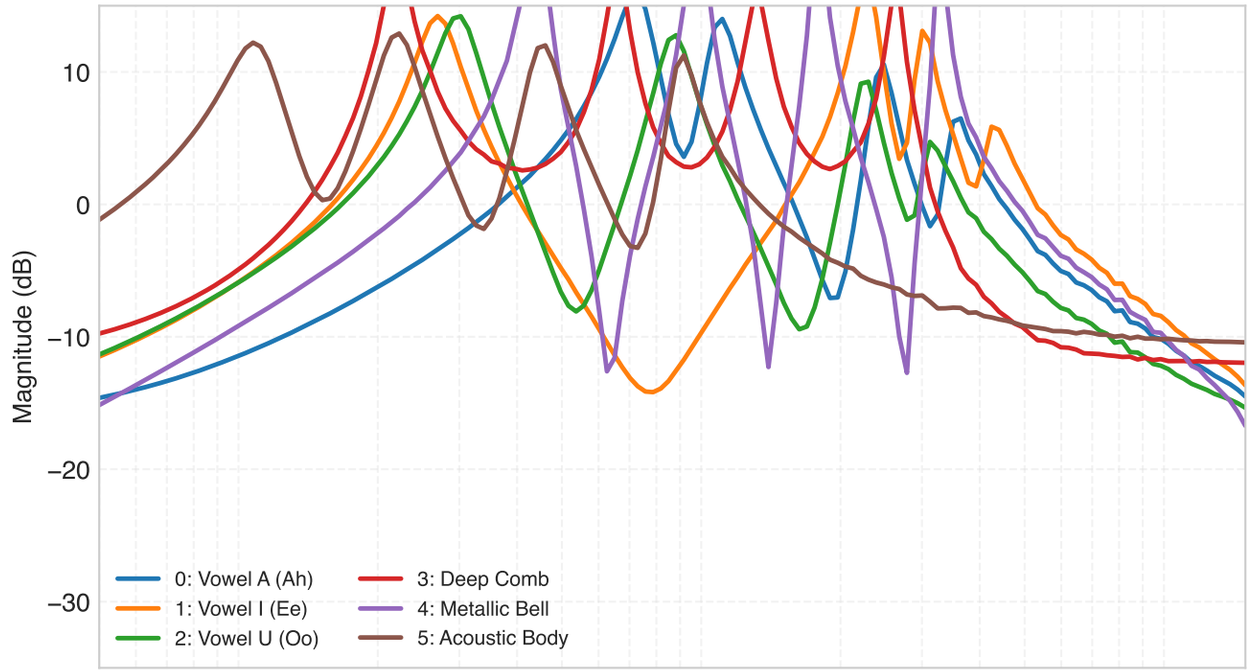


FilterSuite: Advanced Digital Response Morphing (FSZMorph) & All-Pass Dispersion (FSDispersion)

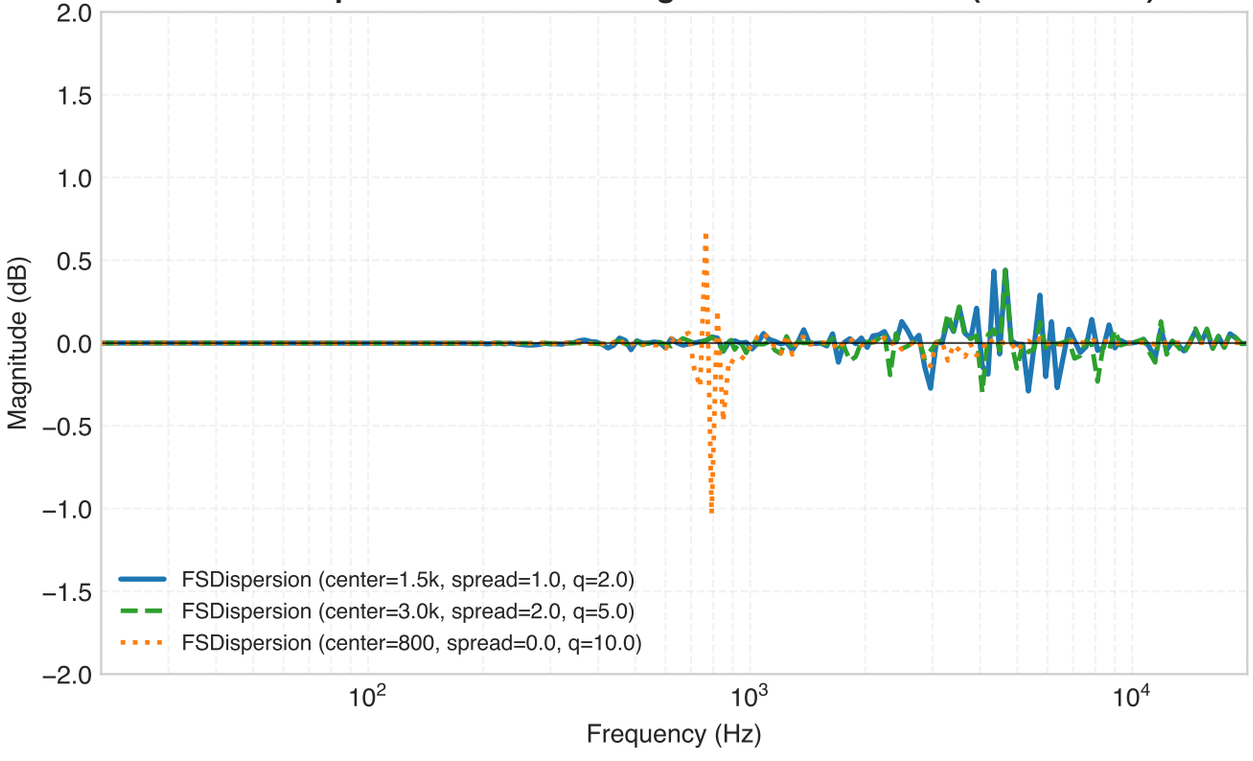
4A. FSZMorph: Continuous Formant Morphing (Vowel A -> Vowel I)



4B. FSZMorph: Spectral Profiles Across All 6 Built-in Presets



4C. FSDispersion: Strict Flat Magnitude Preservation (|H| = 0.0 dB)



4D. FSDispersion: Impulse Temporal Smearing & Parseval Energy Conservation

